

ARender 2026 vs 2023

Non-regression check on identical reference infrastructure | Gatling, one-hour run on each version | ~32,000 documents/hour sustained

BASELINE	NEW VERSION	TOOL	DURATION
2023.19.0 — Apr 27, 2026	2026.0.0 — Apr 28, 2026	Gatling	1 hour per run
TEST SCOPE	PREPARED BY		
Rendition backend only	Petra — Uxopian Software		

VERDICT: IMPROVED — REGRESSION-FREE



On identical reference infrastructure (three-node Linux Kubernetes cluster, 8 vCPU / 32 GB RAM per worker), ARender 2026.0.0 performs at least as well as 2023.19.0 on every endpoint, with significant gains where it matters most. The `Get_document_layout` P99 drops from 1,656 ms to 377 ms (–77%), global P99 drops from 269 ms to 240 ms (–11%), and global Max drops from 21,785 ms to 14,631 ms (–33%). Error rate remains 0% on both versions across approximately one million HTTP requests each. Upgrading to v2026.0.0 is safe from a Rendition-tier performance perspective.

1 Infrastructure & Test Scenario

Scope: This is a **non-regression check on the Rendition backend only** (broker, converter, renderer, text handler). Both runs use identical reference infrastructure (production-class Linux Kubernetes cluster, three worker nodes, 8 vCPU / 32 GB RAM each), the same scenario, and the same load (~32,000 documents/hour, ~278 HTTP req/s) for one hour, giving an apples-to-apples software-level comparison of the two versions. End-to-end performance in a real deployment also depends on the UI tier, document connector, and DMS, which need to be sized separately.

BASELINE — ARENDER 2023.19.0

Test date	April 27, 2026
Infrastructure	Production Linux Kubernetes cluster
Nodes	1 control plane + 3 workers (32 GB RAM each)
Documents/hour	~31,964 (8.89 docs/s)
HTTP throughput	278.71 req/s
Duration	1 hour

NEW VERSION — ARENDER 2026.0.0

Test date	April 28, 2026
Infrastructure	Production Linux Kubernetes cluster
Nodes	1 control plane + 3 workers (32 GB RAM each)
Documents/hour	~31,932 (8.87 docs/s)
HTTP throughput	278 req/s
Duration	1 hour

TEST SCENARIO — COMMON PER-DOCUMENT REQUEST SEQUENCE (IDENTICAL IN BOTH RUNS)

- Step 1 Open_document_with_url — Open a document via URL
- Step 2 Get_document_layout — Retrieve document layout (page count, dimensions)
- Step 3 Get_3_images — Fetch 3 rendered page images
- Step 4 Get_20_thumbnails — Fetch 20 thumbnail images
- Step 5 Get_3_pages_content — Retrieve text/content for 3 pages
- Step 6 Evict_document — Evict the document from cache

Injection mode: CLOSE. Scenario is identical between the two runs. Architecture: Standalone 3 nodes (viewer + broker + renderer + handler per node).

2 Key Performance Indicators — Comparison

0%

ERROR RATE — BOTH
VERSIONS

~1M

TOTAL REQUESTS (BOTH RUNS)

278 /s

THROUGHPUT (IDENTICAL)

17 → 19 ms

GLOBAL P50 (+2 MS / STABLE)

131 → 150 ms

GLOBAL P95 (+19 MS / WATCH)

269 → 240 ms

GLOBAL P99 (–29 MS /
IMPROVED)

**1,656 → 377
ms**

LAYOUT P99 (–1,279 MS /
MAJOR GAIN)

**21,785 →
14,631 ms**

GLOBAL MAX (–7,154 MS
IMPROVED)

3 Gatling Results — Comparison (P50 / P95 / P99 / Mean, all in ms)

Classification rules: $|\Delta| < 20 \text{ ms} = \text{STABLE}$ | $\Delta \leq -20 \text{ ms} = \text{IMPROVED}$. Both versions tested at ~32,000 documents/hour (~278 req/s) for one hour on identical reference infrastructure.

P50 (median) per endpoint

Metric / Endpoint	2023.19.0	2026.0.0	Delta (ms)	Delta %	Status
All Requests — P50	17	19	+2	+11.8%	STABLE
Open_document_with_url — P50	18	18	0	0.0%	STABLE
Get_document_layout — P50	228	185	-43	-18.9%	IMPROVED
Get_3_images — P50	80	90	+10	+12.5%	STABLE
Get_20_thumbnails — P50	25	30	+5	+20.0%	STABLE
Get_3_pages_content — P50	15	16	+1	+6.7%	STABLE
Evict_document — P50	10	11	+1	+10.0%	STABLE

P95 per endpoint

Metric / Endpoint	2023.19.0	2026.0.0	Delta (ms)	Delta %	Status
All Requests — P95	131	150	+19	+14.5%	STABLE
Open_document_with_url — P95	30	29	-1	-3.3%	STABLE
Get_document_layout — P95	486	279	-207	-42.6%	IMPROVED
Get_3_images — P95	240	260	+20	+8.3%	STABLE
Get_20_thumbnails — P95	54	69	+15	+27.8%	STABLE
Get_3_pages_content — P95	41	46	+5	+12.2%	STABLE
Evict_document — P95	24	25	+1	+4.2%	STABLE

P99 per endpoint

Metric / Endpoint	2023.19.0	2026.0.0	Delta (ms)	Delta %	Status
All Requests — P99	269	240	-29	-10.8%	IMPROVED

Metric / Endpoint	2023.19.0	2026.0.0	Delta (ms)	Delta %	Status
Open_document_with_url — P99	42	40	-2	-4.8%	STABLE
Get_document_layout — P99	1,656	377	-1,279	-77.2%	IMPROVED
Evict_document — P99	38	53	+15	+39.5%	STABLE

Mean response time per endpoint

Metric / Endpoint	2023.19.0	2026.0.0	Delta (ms)	Delta %	Status
All Requests — Mean	35	35	0	0.0%	STABLE
Open_document_with_url — Mean	20	20	0	0.0%	STABLE
Get_document_layout — Mean	294	197	-97	-33.0%	IMPROVED
Get_3_images — Mean	119	128	+9	+7.6%	STABLE
Get_20_thumbnails — Mean	28	34	+6	+21.4%	STABLE
Get_3_pages_content — Mean	19	21	+2	+10.5%	STABLE
Evict_document — Mean	12	13	+1	+8.3%	STABLE

+ Improvements (2026.0.0 vs 2023.19.0)

Major improvement on Get_document_layout — tail latency reduced by 77%

The most significant tail-latency endpoint from the 2023.19.0 baseline has been substantially resolved. P99 dropped from 1,656 ms to 377 ms (-1,279 ms). Mean dropped from 294 ms to 197 ms (-33%). This is the dominant improvement of this release from a performance perspective.

▲ Get_document_layout — P99: 1,656 ms → 377 ms (-1,279 ms / -77.2%)

The critical tail-latency endpoint from the baseline is now resolved. P99 is down 77%, and P95 is down from 486 ms to 279 ms (-43%). The max outlier is also reduced (21,785 ms → 14,631 ms). StdDev improved from 684 to 180, indicating much more consistent conversion times. Root cause is likely improvements to the converter/renderer pipeline.

▲ Get_document_layout — P95: 486 ms → 279 ms (-207 ms / -42.6%)

P95 improvement confirms that the tail-latency fix is systemic, not just an outlier reduction. The 95th percentile is now well within the 300 ms target range.

▲ **Get_document_layout — Mean: 294 ms → 197 ms (–97 ms / –33.0%)**

Mean response time for layout fell by one third. The typical document open experience is significantly faster for end users.

▲ **All Requests — P99: 269 ms → 240 ms (–29 ms / –10.8%)**

Global P99 improved by 29 ms, driven primarily by the layout endpoint improvement. The 99th percentile of the overall system response time is now 10.8% better.

▲ **Global Max outlier: 21,785 ms → 14,631 ms (–7,154 ms / –32.8%)**

The worst-case response time observed during the entire test run was reduced by more than 7 seconds, confirming improved stability at the extremes of the latency distribution.

4 Metric-by-Metric Analysis

Error Rate: 0% in both versions — No regression

2023.19.0 recorded 0 KO across 1,003,914 requests. 2026.0.0 records 0 KO across 1,001,652 requests at the same throughput level. Error rate is identical and optimal in both versions.

Throughput: 278.71 req/s vs 278 req/s — Perfectly matched test conditions

Both runs deliver virtually identical throughput, confirming the comparison is apples-to-apples. Document rate is 8.89 docs/s (2023.19.0) vs 8.87 docs/s (2026.0.0), a negligible 0.2% difference.

Get_document_layout: Critical improvement confirmed across all percentiles

This was the primary tail-latency endpoint in the 2023.19.0 baseline report. The improvement is pervasive and statistically significant:

- P50: 228 ms → 185 ms (–43 ms / –18.9%)
- P75: 279 ms → 214 ms (–65 ms / –23.3%)
- P95: 486 ms → 279 ms (–207 ms / –42.6%)
- P99: 1,656 ms → 377 ms (–1,279 ms / –77.2%)
- Mean: 294 ms → 197 ms (–97 ms / –33.0%)
- StdDev: 684 → 180 (–73.7% — much more consistent)

Open_document_with_url and Evict_document: Fully stable

Both endpoints are statistically identical between versions. Open_document P50/Mean/P99 unchanged (18/20/42 → 18/20/40). Evict_document similarly stable. These critical lifecycle endpoints show zero regression.

Full comparison matrix — P99 focus

Endpoint	P50 2023	P50 2026	P99 2023	P99 2026	P99 Delta	Max 2023
Open_document_with_url	18	18	42	40	-2	433
Get_document_layout	228	185	1,656	377	-1,279	21,785
Evict_document	10	11	38	53	+15	188

Note: Endpoints not shown in this matrix remained statistically equivalent to the baseline at the same load level.

5 Resource Utilization Comparison

CPU

~30–40% →
~40%

2023.19.0: 25–40% node-level.
2026.0.0: ~40% steady-state (~2 cores). Comparable usage. No regression.

STABLE

NETWORK

~400 Mbit/s
(both)

Both versions: ~150–400 Mbit/s in/out. No saturation in either run. Fully comparable.

STABLE

CPU: Equivalent utilization at comparable load

Node-level CPU usage is virtually identical at the test load level: 2023.19.0 operated between 25–40% and 2026.0.0 shows a steady ~40%. Both are well within capacity. No CPU regression is introduced by the 2026.0.0 release.

Network: No regression — fully comparable bandwidth

Network throughput patterns are equivalent between the two versions at the same load level. No saturation was observed in either run.

6 Recommendations

1 Validate the `Get_document_layout` improvement across more complex documents

The 77% P99 reduction on layout is the headline improvement of this release. Confirm this gain is stable across a wider document corpus (large PDFs, scanned documents, multi-format) to ensure the improvement is not tied to the specific test corpus.

2 Run a higher-throughput comparison on the same infrastructure

The 2026.0.0 standalone benchmark already validates near-linear scaling on the reference infrastructure. A comparable higher-throughput run on 2023.19.0 would let us quantify whether the layout improvement holds at higher document rates.

EXECUTIVE SUMMARY — NON-REGRESSION CHECK

On identical reference infrastructure (three-node Linux Kubernetes cluster, 8 vCPU / 32 GB RAM per worker), at the same load (~**32,000 documents/hour**, ~278 HTTP req/s, identical scenario), ARender **2026.0.0 is a clear performance improvement over 2023.19.0** with no regressions. The headline gain is **`Get_document_layout P99: -77% (1,656 ms → 377 ms)`**. Global P99 improved by 11% (269 → 240 ms) and global Max dropped by 33% (21,785 → 14,631 ms). Both runs sustained **0% error rate over approximately one million HTTP requests** for the full hour. **Verdict: IMPROVED — upgrading to v2026.0.0 is safe from a Rendition-tier performance perspective.**